

03:24

Problem Solving – Definition... *agent world → environment*

- Here search space or problem space is abstract space which has all valid states that can be generated by the application of any combination of operators or objects.
- A problem space can have one or more **solutions**.
- Solution is a combination of objects and operations that achieve the goals.
- Search refers a search of solution in problem space.

05:39

Goal Based Agent or Problem Solving Agent

- Goal-based agents are usually called **planning agents**.
- Problem solving begins with the definitions of problems and their solutions.
- We then describe several general-purpose search algorithms that can be used to solve these problems.
- We will see several uninformed search algorithms— algorithms that are given no information about the problem other than its definition.
- **Informed search algorithms**, on the other hand, can do quite well given some guidance on where to look for solutions

*BFS
DFS
UCS
A**

05:41

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06:23

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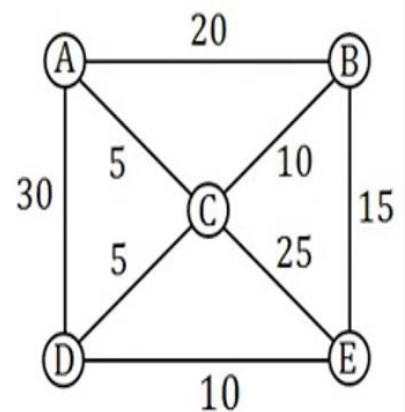
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G.B.F.
A*
A²*

06:38

Problem Solving Agent

- **Problem-solving agents:** used to find sequence of actions that achieve goals.
- An example,
- Each node represents a city, and the cost to travel from a city to another is denoted by the number over the edge connecting the nodes of those 2 cities.
- In order for an agent to solve a problem it should pass by 2 phases of formulation:



08:42

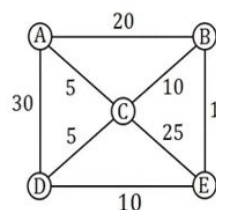
Goal Formulation and Problem Formulation

Goal Formulation:

- Problem solving is about having a goal we want to reach, (i.e. I want to travel from 'A' to 'E').
- Goal is a set of environment states in which the goal is satisfied.

Problem Formulation:

- A problem formulation is about deciding what actions and states to consider,
- describe states as "in (CITYNAME)" where CITYNAME is the name of the city in which we are currently in.



- The process of finding such sequence is called **search**, a search algorithm is like a black box which takes **problem** as input returns a **solution**, and once the solution is found the **sequence of actions** carried out, and this is called the **execution phase**.

09:06

Goal Formulation and Problem Formulation

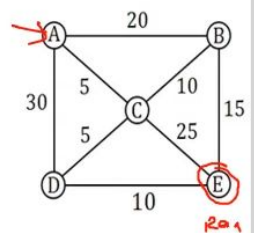
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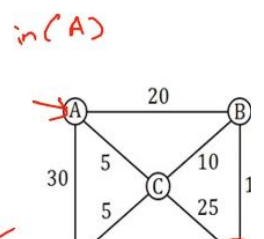
11:02

Goal Formulation and Problem Formulation

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11:22

Formulating Problems

- A problem can be defined formally by 4 components:

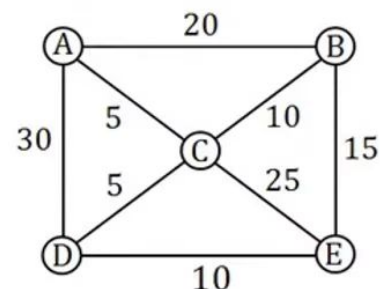
- **Initial state, successor function, goal test, and path cost.**

1. Initial State:

- Specify one or more states that describes the possible situations from which the problem solving process starts.

2. Successor Function :

- a description of the possible actions available to the agent, it is common to describe it by means of a successor function, given state x then $SUCCESSOR-FN(x)$ returns a set of ordered pairs
- **<action, successor>** where action is a legal action from state x and successor is the state in which we can be by applying action.
- The initial state and the successor function together defined what is called state space which is the set of all possible states reachable from the initial state {e.g: in(A), in(B), in(C), in(D), in(E)}.

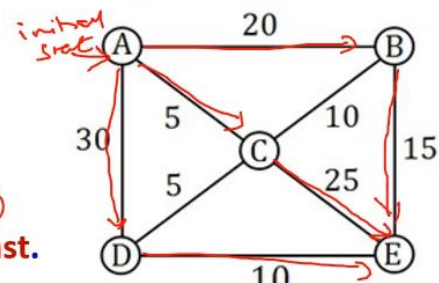


14:11

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14:16

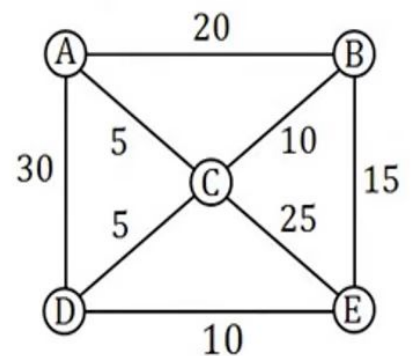
Formulating Problems...

3. Goal Test:

- Decide whether the current state is a goal state
- {i.e.: is the current state is $in(E)$?

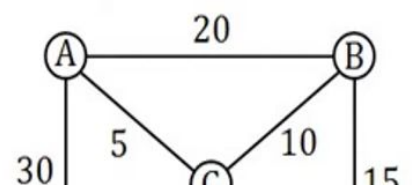
4. Path cost:

- a function that assigns a numeric value to each path, $C(A, \text{cost}, B)$
 - each step we take in solving the problem should be somehow weighted,
 - If I travel from A to E our agent will pass by many cities, the cost to travel between two consecutive cities should have some cost measure, {e.g: Traveling from 'A' to 'B' costs 20 km or it can be typed as $c(A, 20, B)$ }.
- A solution to a problem is path from the initial state to a goal state,
 - and solution quality is measured by the path cost, and the optimal solution has the lowest path cost among all possible solutions.



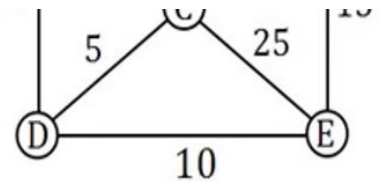
14:52

Formulating Problems...



3. Goal test: ✓ *knowledge about the goal*

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15:22

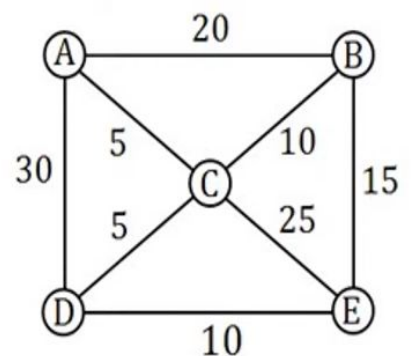
Formulating Problems...

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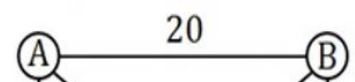
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15:26

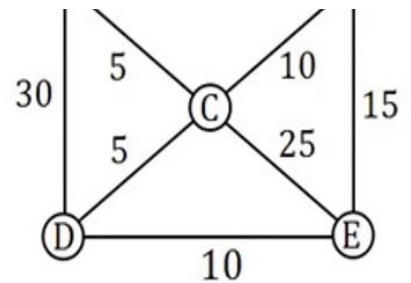


Formulating Problems...

3. Goal Test: ✓

knowledge about the goal

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17:39

Thank You

• Problem Formulation

Introduction

- Problem Solving – Definition ✓
- Goal Based Agent / Problem Solving Agent ✓
- Goal Formulation and Problem Formulation ✓
- Formulating Problems ✓
 - Initial State, Successor Function, Goal Test and Path Cost.
- Example problems – Vacuum world problem, 8 puzzle problem, 8 Queen problem, and missionaries and cannibals problem

01:10

Problem Solving - Definition

- Problem solving is a process of generating solutions from observed data.)
- Problem is characterized by a set of goals, set of objects, and set of operations.
- The method of solving problem through AI involves the process of
 - defining the search space,
 - deciding start and goal states
 - finding the path from start state to goal state through search space.

02:47

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